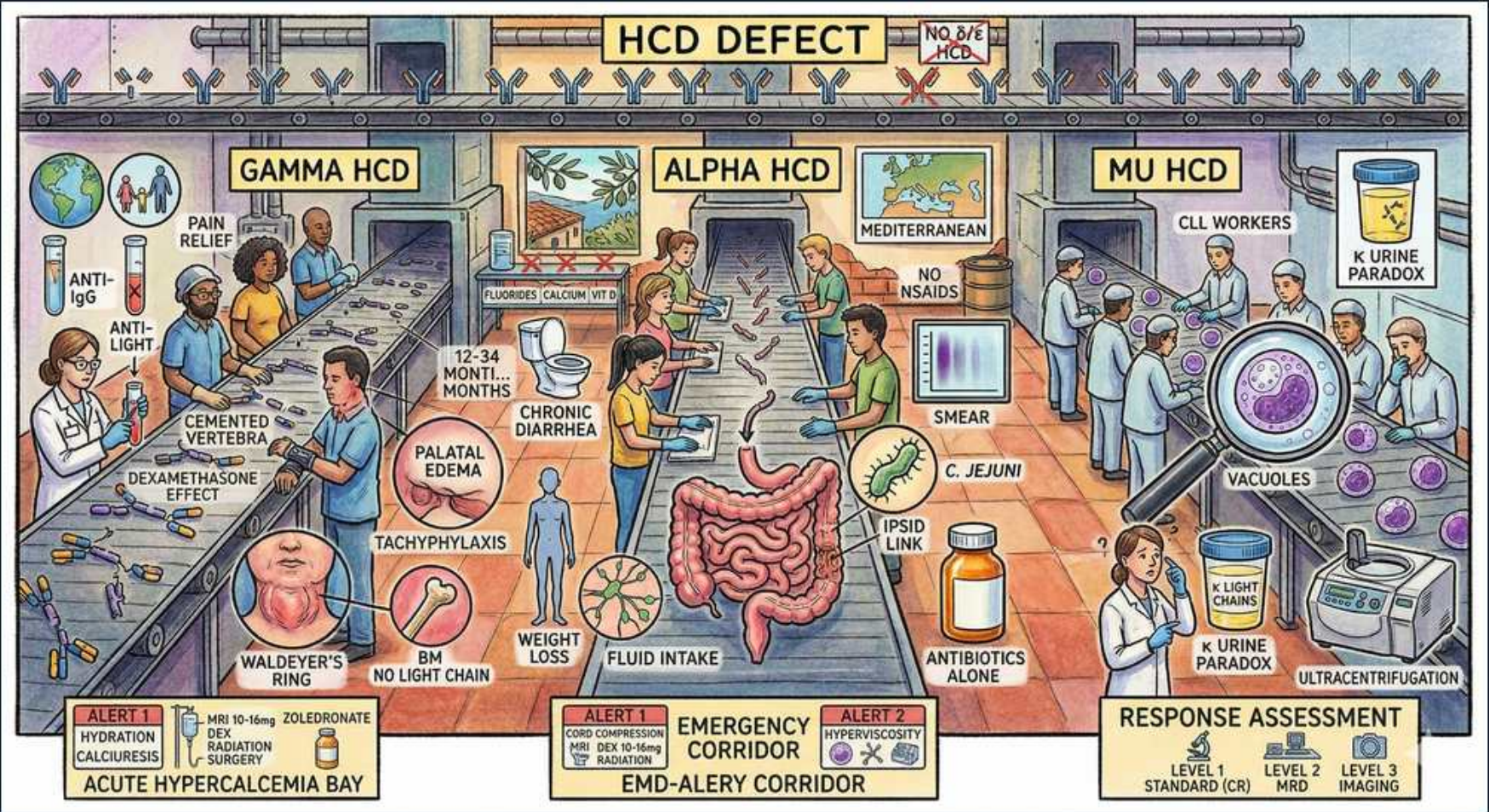


Plasma Cell — Heavy Chain Diseases (Gamma/Alpha/Mu)



1. HCD defect

WHAT YOU SEE

The HCD DEFECT factory header stamped 'NO LIGHT CHAIN' — heavy chain diseases churn out a truncated heavy chain that can't grab a light chain, the defining flaw of all three types.

WHAT IT ENCODES

Heavy chain diseases (HCDs) are rare B-cell lymphoproliferative disorders in which a clone secretes a TRUNCATED, incomplete immunoglobulin heavy chain that CANNOT bind a light chain. The deleted portion is usually part of the variable/CH1 region, so the defective heavy chain assembles no light chain. Three types are named by isotype: gamma (Franklin), alpha (Seligmann, most common), and mu (rarest).

BOARD TRAP

WRONG to expect a normal intact monoclonal protein on routine SPEP. The truncated heavy chain often migrates as a broad/abnormal band, and the defining clue is an isolated heavy chain with NO light chain on immunofixation/immunoselection. Discriminator: absence of an associated light chain across all HCD types separates them from typical myeloma/MGUS (which have light chains).

2. Gamma HCD

WHAT YOU SEE

The GAMMA HCD assembly line — gamma (IgG) HCD is Franklin disease, a lymphoplasmacytic-lymphoma-like process with nodes and spleen, not lytic bone.

WHAT IT ENCODES

Gamma (IgG) heavy chain disease = Franklin disease, a heterogeneous lymphoplasmacytic disorder resembling lymphoplasmacytic lymphoma. Presents with constitutional symptoms, lymphadenopathy, hepatosplenomegaly, and anemia; course ranges from indolent to aggressive lymphoma. Treat the underlying lymphoproliferative disease (chemo/immunotherapy) when symptomatic.

BOARD TRAP

WRONG to mistake gamma HCD for multiple myeloma because of an IgG band. Franklin disease is a lymphoplasmacytic lymphoma-like process (nodes, spleen, Waldeyer ring), not a lytic-bone marrow plasma-cell tumor, and the heavy chain lacks a light chain. Discriminator: lymphadenopathy/palatal edema + truncated IgG with no light chain = Franklin gamma HCD, not myeloma.

3. Gamma palatal edema

WHAT YOU SEE

A swollen PALATAL EDEMA / Waldeyer's ring figure on the gamma line — lymphoid infiltration of Waldeyer's ring puffs the palate/uvula, the signature exam buzzword for Franklin disease.

WHAT IT ENCODES

Gamma HCD classically causes palatal/uvular edema due to lymphoid infiltration of Waldeyer's ring (tonsillar/pharyngeal lymphoid tissue), which can threaten the airway. This is the signature physical-exam buzzword for Franklin disease.

BOARD TRAP

WRONG to attribute palatal/uvular edema in this setting to angioedema or infection. In a patient with lymphadenopathy and an abnormal heavy-chain band, palatal edema is the Waldeyer-ring clue to gamma (Franklin) HCD. Discriminator: palatal edema + Waldeyer ring + truncated IgG = gamma HCD.

4. Gamma autoimmune link

WHAT YOU SEE

ANTI-IgG / anti-light-chain autoantibody icons on the gamma line — gamma HCD rides with autoimmune disease (rheumatoid arthritis, Sjogren), which drives the clone.

WHAT IT ENCODES

Gamma HCD is associated with autoimmune disease, most notably rheumatoid arthritis and Sjögren syndrome; HCD can be diagnosed during workup of an autoimmune patient. The chronic immune stimulation is thought to drive the clonal expansion.

BOARD TRAP

WRONG to assume a monoclonal heavy chain in a patient with rheumatoid arthritis is merely a reactive paraprotein. Gamma HCD has a recognized association with autoimmune disease and can be missed. Discriminator: a new abnormal heavy-chain band in an RA/Sjögren patient warrants immunofixation to look for HCD.

5. Alpha HCD

WHAT YOU SEE

The ALPHA HCD assembly line — alpha (IgA) HCD is Seligmann disease, the most common HCD and the secretory face of small-bowel IPSID MALT lymphoma.

WHAT IT ENCODES

Alpha (IgA) heavy chain disease = Seligmann disease, the MOST COMMON HCD. It is essentially the secretory phenotype of immunoproliferative small-intestinal disease (IPSID), a MALT-type lymphoma of the proximal small bowel. The clone infiltrates intestinal lamina propria and mesenteric nodes.

BOARD TRAP

WRONG to confuse alpha HCD demographics/site with gamma HCD. Alpha HCD targets the GI tract (small bowel/mesenteric nodes) in young Mediterranean/Middle-Eastern adults, whereas gamma HCD targets systemic lymphoid tissue/Waldeyer ring. Discriminator: GI/IPSID + IgA truncated heavy chain = alpha (Seligmann); nodes/palate + IgG = gamma (Franklin).

6. Mediterranean / young

WHAT YOU SEE

MEDITERRANEAN seaside scenery on the alpha line — alpha HCD/IPSID strikes YOUNG adults of Mediterranean/Middle-Eastern origin (high-yield for an Israeli board).

WHAT IT ENCODES

Alpha HCD/IPSID classically affects young adults (often 2nd–3rd decade) of Mediterranean, Middle-Eastern, and North African origin, in areas of poor sanitation and chronic enteric infection — highly relevant epidemiology for a Middle-Eastern/Israeli board exam.

BOARD TRAP

WRONG to anchor on elderly-patient assumptions used for myeloma. Alpha HCD strikes YOUNG adults in endemic Mediterranean/Middle-Eastern regions. Discriminator: a young Mediterranean patient with chronic diarrhea/malabsorption and an IgA abnormality should raise IPSID/alpha HCD, not classic myeloma.

7. GI malabsorption

WHAT YOU SEE

An inflamed INTESTINE with chronic diarrhea and weight loss on the alpha line — diffuse small-bowel lymphoplasmacytic infiltration causes the malabsorption of IPSID/alpha HCD.

WHAT IT ENCODES

Alpha HCD = IPSID presents with chronic diarrhea, steatorrhea, malabsorption, abdominal pain, weight loss, and finger clubbing from diffuse small-bowel lymphoplasmacytic infiltration. Endoscopic biopsy shows dense lamina-propria plasma-cell/lymphoid infiltrate; the truncated alpha heavy chain may be detected in serum and intestinal secretions.

BOARD TRAP

WRONG to chalk up the malabsorption to celiac or tropical sprue without considering IPSID in an endemic young patient. Discriminator: chronic malabsorption in a young Mediterranean adult plus an IgA heavy-chain abnormality and dense plasma-cell small-bowel infiltrate = IPSID/alpha HCD; small-bowel biopsy with immunofixation distinguishes it.

8. C. jejuni / antibiotics

WHAT YOU SEE

A C. JEJUNI bug beside an 'antibiotics alone' box — early antigen-driven IPSID (like H. pylori gastric MALT) can regress with antibiotics alone before it transforms.

WHAT IT ENCODES

Early-stage IPSID is antigen-driven and linked to Campylobacter jejuni infection. Like H. pylori-driven gastric MALT lymphoma, EARLY IPSID can regress with antibiotics alone (e.g., tetracycline) — eradicating the antigenic stimulus. Advanced disease that has progressed to overt lymphoma requires chemotherapy.

BOARD TRAP

WRONG to give chemotherapy to early-stage alpha HCD/IPSID, or to give antibiotics to advanced/overt lymphoma expecting a response. Discriminator: early antigen-driven IPSID (analogous to H. pylori MALT) responds to antibiotics alone; only transformed/advanced disease needs chemo — stage determines treatment.

9. Mu HCD

WHAT YOU SEE

The MU HCD assembly line staffed by CLL WORKERS — mu (IgM) HCD is the rarest, arising on a background of CLL/chronic lymphoproliferative disease.

WHAT IT ENCODES

Mu (IgM) heavy chain disease is the RAREST HCD, typically arising in the setting of chronic lymphocytic leukemia (CLL) or another chronic lymphoproliferative disorder. Course mirrors the underlying CLL/lymphoma; hepatosplenomegaly is common while peripheral lymphadenopathy is often absent.

BOARD TRAP

WRONG to expect lymphadenopathy as in gamma HCD. Mu HCD usually presents with hepatosplenomegaly and a CLL-like marrow but WITHOUT prominent peripheral nodes. Discriminator: rarest HCD + CLL/lymphoproliferative background + vacuolated marrow plasma cells = mu HCD.

10. Mu vacuolated cells

WHAT YOU SEE

VACUOLATED plasma cells drawn in the mu marrow — bubbly vacuolated marrow plasma cells are the morphologic hallmark of mu HCD.

WHAT IT ENCODES

Mu HCD shows characteristic VACUOLATED plasma cells in the bone marrow — a morphologic hallmark on aspirate. Combined with a CLL-like lymphocytosis, these vacuolated plasma cells point to the diagnosis.

BOARD TRAP

WRONG to overlook vacuolated marrow plasma cells as nonspecific. In a CLL-like picture, vacuolated plasma cells are the morphologic clue to mu HCD. Discriminator: vacuolated marrow plasma cells + CLL background + IgM heavy chain without light chain in serum = mu HCD.

11. Mu kappa light chain in urine

WHAT YOU SEE

The KAPPA URINE PARADOX panel on the mu line — mu HCD paradoxically spills free kappa light chains into the urine even though the serum heavy chain is light-chain-less.

WHAT IT ENCODES

Mu HCD paradoxically DOES often produce free kappa light chains that appear in the urine (Bence-Jones proteinuria), even though the serum defect is an isolated, truncated mu heavy chain that cannot assemble with light chains. The light chains are made but not incorporated into intact Ig.

BOARD TRAP

WRONG to exclude a 'heavy chain disease' because the patient has urinary kappa light chains. The kappa-urine paradox is specifically characteristic of mu HCD. Discriminator: isolated serum mu heavy chain (no light chain) PLUS free kappa in the urine = mu HCD, not light-chain myeloma.

12. Ultracentrifugation/IFE

WHAT YOU SEE

ULTRACENTRIFUGATION + immunofixation bench in the response corner — diagnosis needs these specialized studies to show a heavy chain with NO light chain (plain SPEP misses it).

WHAT IT ENCODES

Definitive diagnosis of any HCD requires demonstrating an isolated heavy chain with NO associated light chain — done by serum/urine immunofixation electrophoresis and immunoselection (specialized techniques like immunoselection or ultracentrifugation/gel filtration to isolate the abnormal heavy chain). Routine SPEP alone is often inadequate because the truncated protein may not form a typical sharp spike.

BOARD TRAP

WRONG to rule out HCD on a normal or merely 'abnormal broad band' SPEP. Discriminator: HCD diagnosis hinges on immunofixation/immunoselection showing heavy chain WITHOUT light chain — order these specialized studies rather than relying on SPEP appearance.

13. No NSAIDs / supportive

WHAT YOU SEE

A 'NO NSAIDs' sign with fluoride/calcium/vitamin D supplies — avoid nephrotoxic NSAIDs in these paraprotein-stressed kidneys and give supportive bone/nutrition care.

WHAT IT ENCODES

Management is treating the underlying lymphoproliferative disorder (antibiotics for early IPSID; chemoimmunotherapy for advanced/aggressive disease) plus supportive care. Avoid nephrotoxic NSAIDs (paraprotein-associated renal vulnerability) and support bone/nutritional health (calcium, vitamin D) and malabsorption.

BOARD TRAP

WRONG to manage HCD with NSAIDs for pain in a paraproteinemic patient — NSAIDs add nephrotoxic insult to paraprotein-stressed kidneys. Discriminator: treat the underlying clone and use NSAID-sparing analgesia; supportive care addresses malabsorption, bone, and renal protection.